

1) Dimensional analysis! Let the units for mass, length and time be $\mathcal{M}, \mathcal{L}, \mathcal{T}$ respectively. Then the units for the Planck units are:

- $[R] = [1 \text{ m}] = \mathcal{L}$
- $[\rho] = [1 \text{ kg} \cdot \text{m}^{-3}] = \mathcal{M} \mathcal{L}^{-3}$
- $[E] = [1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}] = \mathcal{M} \mathcal{L}^2 \mathcal{T}^{-2}$
- $[t] = [1 \text{ s}] = \mathcal{T}$

$$R = \chi \rho^\alpha E^\beta t^\gamma$$

equating:

$$\begin{cases} \alpha + \beta = 0 \\ -3\alpha + 2\beta = 1 \\ -2\beta + \gamma = 0 \end{cases}$$

which gives

$$R = \chi \rho^{-1/5} E^{1/5} t^{2/5} = \chi \sqrt[5]{Et^2/\rho}$$

The question allows us to assume $\chi = 1$.

Measuring the diagram: $R = \frac{3.30 \text{ cm}}{2.00 \text{ cm}} \cdot 100 \text{ m} = 165 \text{ m}$

$$E = \frac{\rho R^5}{t^2} = \frac{1.2 \cdot 165^5}{(25 \times 10^{-3})^2} = 2.348 \times 10^{14} \text{ J}$$

which is equivalent to $(2.348 \times 10^{14}) / (4.184 \times 10^9) = \mathbf{56000 \text{ tonnes of TNT}}$.

2) The adiabatic invariant is the area of the orbit in phase-space coordinates. This means that when plotting the momentum-position graph of the pendulum, the area of the region (in this case, an ellipse) will be constant when the system is perturbed slowly. (The reason for this is quantum mechanical; phase space A can only take on certain integer/half-integer multiples of the Planck constant h , and slow perturbations do not change the "quantum state" of the system.)

$$A = \pi s_{\max} p_{\max}$$

Here, $s_{\max} = L\theta$, and by conservation of energy,

$$mgL(1 - \cos \theta) \approx mgL \frac{\theta^2}{2} = \frac{1}{2}mv^2$$

$$v = \sqrt{gL\theta^2}$$

$$\text{so } A = \pi s_{\max} p_{\max} = \pi L\theta \cdot mv = \pi m \sqrt{g} \left[L^{\frac{3}{2}} \cdot \theta^2 \right]$$

So $L^{\frac{3}{2}} \cdot \theta^2$ is constant, which means as L is halved, $\theta_{\text{new}} = \theta \cdot 2^{\frac{3}{4}}$, so the physical amplitude of oscillations $s_{\text{new}} = s_{\text{old}} \cdot 2^{-\frac{1}{4}}$.

3) The time of flight is related to the velocity and height by

$$h = vt + \frac{1}{2}gt^2$$

We thus know (**downward positive**)

$$h = v_{1y}t_1 + \frac{1}{2}gt_1^2 = v_{2y}t_2 + \frac{1}{2}gt_2^2$$

and by conservation of momentum at the top of the flight,

$$Mv_{1y} + Mv_{2y} + Mv_{2y} = 0 \therefore v_{1y} + 2v_{2y} = 0$$

[The fragments are in general not 120 degrees apart. They only need to satisfy conservation of momentum.]

$$h = v_{1y}t_1 + \frac{1}{2}gt_1^2, h = -\frac{v_{1y}}{2}t_2 + \frac{1}{2}gt_2^2$$

By substituting the $v_{1y} = \frac{h - \frac{1}{2}gt_1^2}{t_1}$ from the first equation into the second, the

final answer is $h = \frac{gt_1t_2(t_1+2t_2)}{2(2t_1+t_2)}$.

4) (i) At the center, the positive charge is distributed spherically symmetrically around the electron and thus the net force exerted by the Coulomb forces must be 0 (by symmetry, it must have equal magnitude in all directions). Thus the electron is equilibrium in the center.

When displaced, it experiences a net force due to the positive charge contained in the sphere of radius r , which is $q_+ = er^3/R^3$. The restoring force is thus

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_+(-e)}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{(er^3/R^3)(-e)}{r^2} = -\frac{e^2}{4\pi\epsilon_0 R^3} r$$

(ii) $F = -Kr = m_e \ddot{r} \therefore \ddot{r} + (K/m_e)r = 0$

$$\omega = \sqrt{K/m_e} = \sqrt{\frac{e^2}{4\pi\epsilon_0 R^3 m_e}}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{e^2}{4\pi\epsilon_0 R^3 m_e}}$$

This can be simplified further into $f = \frac{e}{4\pi} \sqrt{\frac{1}{\pi\epsilon_0 R^3 m_e}}$

(iii) Make R the subject, $R = \left[\frac{e^2}{16\pi^3 f^2 \epsilon_0 m_e} \right]^{\frac{1}{3}} = 1.016 \times 10^{-10} \text{ m} \approx 1.02 \times 10^{-10} \text{ m}$

5) (i) Total energy of the system is $E_0 = M_0 c^2$. By conservation of energy,

$$E_0 = M_0 c^2 = E_{rad} + \gamma m c^2$$

(where it is given that $m = f M_0$).

(ii) Conservation of momentum:

$$0 = p_{rad} + p_{ship} = -\frac{E_{rad}}{c} + \gamma m v c$$

(Note in this question v is the speed relative to light).

(iii) Substituting second equation into first:

$$M_0 c^2 = \gamma m v c^2 + \gamma m c^2 = (\gamma + \gamma v) f M_0 c^2 \therefore \gamma f + \gamma v f = 1$$

(iv) $\gamma v = \frac{v}{\sqrt{1-v^2}} = \sqrt{\frac{v^2}{1-v^2}} = \sqrt{\frac{1}{1-v^2} - 1} = \sqrt{\gamma^2 - 1}$. Substituting into the third equation:

$$\begin{aligned} (\gamma + \sqrt{\gamma^2 - 1}) f &= 1 \therefore f = \frac{1}{\gamma + \sqrt{\gamma^2 - 1}} = \gamma - \sqrt{\gamma^2 - 1} \\ \therefore f^2 - 2\gamma f + 1 &= (\gamma - \sqrt{\gamma^2 - 1})^2 - 2\gamma(\gamma - \sqrt{\gamma^2 - 1}) + 1 \\ &= \gamma^2 + \gamma^2 - 1 - 2\gamma\sqrt{\gamma^2 - 1} - 2\gamma^2 + 2\gamma\sqrt{\gamma^2 - 1} + 1 = 0 \end{aligned}$$

(v) $f = 10 - \sqrt{10^2 - 1} = 10 - \sqrt{99} = 0.0501$. It is indeed possible to construct a spaceship with this f .

(vi) Mass of fuel consumed is $M_{fuel} = M_0 - m = (1 - f)M_0 = 0.9499M_0$ and $E_{rad} = \gamma f M_0 v c^2 = 0.4987M_0 c^2$. Thus $M_{fuel} c^2 > E_{rad}$.

The mathematical explanation is that

$$E_{rad} = M_0 c^2 - \gamma m c^2 < (M_0 - m) c^2 = M_{fuel} c^2$$

The physical explanation is that when the rocket fuel is converted into photons, energy is transferred to the spacecraft to propel it in one direction. This lost energy causes the total energy of emitted radiation to be less than the consumed fuel.

6) The induced electric field E at the point of the ring satisfies

$$\oint \vec{E} \cdot d\vec{l} = E \cdot 2\pi R = \varepsilon = A \frac{dB}{dt} = \pi R^2 \alpha \therefore E = R\alpha/2$$

The (tangential) force is $F = qE = qR\alpha/2$ which leads to constant tangential acceleration $a_T = F/m = qR\alpha/2m$. The speed of the small ring is thus $v = a_T t = qR\alpha t/2m$ and the force it exerts on the big ring is (arguing by centrifugal force, or by centripetal force + Newton's 3rd Law)

$$F_c = mv^2/R = \frac{q^2 \alpha^2 R}{4m} t^2$$

The small ring moves with constant angular acceleration and moves faster and faster around the ring.

The small ring also experiences magnetic force $F_B = qvB = \frac{q^2 R \alpha t}{2m} (B_0 + \alpha t)$ towards the centre of the circle. Hence the normal force of the big ring on the small ring:

$$N = F_c - F_B = -\frac{q^2 R \alpha B_0}{2m} t - \frac{q^2 \alpha^2 R}{4m} t^2$$

Hence the normal force of the small ring on the big ring:

$$-N = \frac{q^2 R \alpha B_0}{2m} t + \frac{q^2 \alpha^2 R}{4m} t^2$$

7) (i) $I = I_0(\cos^2 20.0^\circ)^3 = 6.885$

(ii) $I = I_0(\cos^2 30.0^\circ)^2 = 5.625$

(iii) The angle is in fact $\theta_2 = 30.0^\circ$. To show this, we know

$$I = I_0 \cos^2 \theta_2 \cos^2(\theta_3 - \theta_2)$$

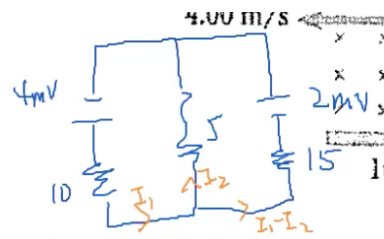
and we want to maximise $|\cos \theta_2 \cos(\theta_3 - \theta_2)|$. One way to find the maximum is to differentiate with respect to θ_2 and equate it to 0; another way is to note

$$|\cos \theta_2 \cos(\theta_3 - \theta_2)| = \left| \frac{\cos \theta_3 + \cos(\theta_3 - 2\theta_2)}{2} \right|$$

which reaches maximum when $\theta_3 - 2\theta_2 = 0$.

8) For one rod, $\varepsilon = Blv$, thus $\varepsilon_1 = 4mV$, $\varepsilon_2 = 2mV$. Solving the circuit below:

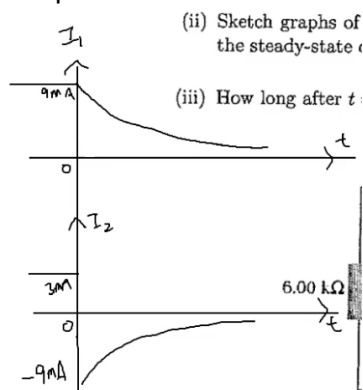
$$\begin{aligned} 4 - 10I_1 - 5I_2 &= 0 \\ 2 + 5I_2 - 15(I_1 - I_2) &= 0 \\ I_1 &= \frac{18}{55}mA, I_2 = \frac{8}{55}mA \end{aligned}$$



9) The loop containing R_2 and the loop containing R_1 and L can be treated as independent. An inductor preserves the current when the switch is opened, thus it will push 9mA of current through both resistors.

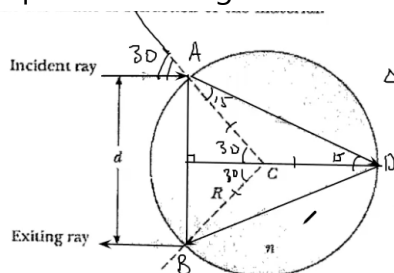
(i) Initially, inductor acts as closed circuit. Thus the current is $I = \frac{V}{R} = \frac{18V}{2k\Omega} = 9mA$. $\mathcal{E}_L(t = 0^+) = IR = 9mA \cdot (6 + 2)k\Omega = 72V$, with point **b** having higher potential than **a** (opposite to the battery).

(ii) R_1 current begins at 9mA, then decay. The equation: $I_1 = I_L e^{-\frac{R}{L}t}$, $I_L = 9mA$. R_2 current was 3mA, then decaying from -9mA due to the definition of direction of current. $I_2 = -I_L e^{-\frac{R}{L}t}$. Here, the R is the effective resistance from the point of view of the inductor, thus $R = 2 + 6 = 8k\Omega$.



(iii) $2 = 9e^{-\frac{8k\Omega}{0.400H}t}$, $\frac{8k\Omega}{0.400H}t = \ln \frac{9.00}{2.00} \therefore t = 7.52 \times 10^{-5}s$

10) Because $d = R$, we can easily find the angles (in degrees), due to equilateral triangle ABC.



and $n = \frac{\sin 30^\circ}{\sin 15^\circ} = \mathbf{1.93}$.